2024, Mar. 2024 Release LOC:Acute Adult
Antepartum

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Instruction: This subset addresses pregnant individuals who require hospital management due to complications that develop during the antepartum period. Evaluation and management of the antepartum pregnant individual should be individualized, based on the type and severity of presenting symptoms, risk factors, concurrent conditions, barriers to discharge, and social determinants of health, and include shared decision-making. Chronic hypertension, gestational hypertension or preeclampsia, eclampsia, and hemolysis, elevated liver enzymes, and low platelet count (HELLP) syndrome are excluded and can be found in the Hypertensive Disorders of Pregnancy subset. Utilize the Labor and Delivery subset for prelabor rupture of membranes, labor induction, and when delivery is indicated.

Criteria for the following antepartum risk factors and conditions are found within this subset:
- Decreased fetal movement or fetal compromise
- High risk of fetal compromise:
    - Abnormal antepartum surveillance testing (i.e., amniotic fluid index, biophysical profile, nonstress test, single deepest vertical pocket, stress test)
    - Chorioamniotic separation (CAS)
    - Fetal anemia
    - Fetal growth restriction with absent or reversed end-diastolic flow
    - Fetal growth restriction with oligohydramnios
    - Monochorionic-monoamniotic twins and expectant management
    - Multiple gestation complications (i.e., fetal growth restriction, growth discordance, single fetal demise of one twin, twin anemia polycythemia sequence (TAPS), twin-to-twin transfusion syndrome (TTTS), twin reversed arterial perfusion (TRAP) sequence, umbilical cord entanglement or compression)
    - Prolapse of amniotic sac
- Nausea and vomiting of pregnancy or hyperemesis gravidarum
- Placental complications:
    - Placental abruption
    - Placenta previa
    - Placenta accreta spectrum (i.e., placenta accreta, placenta increta, placenta percreta)
    - Vasa previa
- Post cerclage
- Pregestational and gestational diabetes mellitus

- Preterm labor
- Preterm prelabor rupture of membranes
- Vaginal bleeding

Evaluation and Treatment:
The degree of maternal or fetal stability and the risk-benefit ratio of prolonging pregnancy for fetal maturation influence the decision to proceed with delivery. For pregnant individuals in whom pregnancy prolongation is indicated, continued maternal and fetal surveillance may take place in the inpatient or outpatient setting for specific conditions and is dependent on a multitude of factors that may include, but are not limited to, the risk for rapid destabilization and the ability and resources to comply with outpatient recommendations. Inpatient management of the antepartum patient may include fetal and maternal monitoring and interventions to stabilize the underlying condition.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:

- ⊖ **Episode Day 1, One:** [1, 2]
  (Symptom or finding within 24h)
  - ⊖ **OBSERVATION, ≥ One:** [3]
    - ⊖ Decreased fetal movement and, **Both:** [4]
      - Fetal heart rate monitoring [5]
      - Ultrasound assessment [6]
    - ⊖ Labor or preterm labor rule out and, **Both:** [7, 8]
      - Fetal heart rate monitoring [5]
      - Uterine monitoring [9]
    - ⊖ Persistent nausea and vomiting of pregnancy or hyperemesis gravidarum and, ≥ **One:** [10, 11]
      - Antiemetic ≥ 2x/24h [12]
      - IV fluid [13]
    - ⊖ Post cerclage and, **Both:** [14]
      - Fetal heart rate monitoring [5]
      - Uterine monitoring [9]
    - ⊖ Pregestational or gestational diabetes mellitus and, **All:** [15, 16, 17]
      - ⊖ Blood glucose, ≥ **One:** [18]
        - Fasting ≥ 95 mg/dL(5.3 mmol/L)
        - 1h postprandial ≥ 140 mg/dL(7.8 mmol/L) or 2h postprandial ≥ 120 mg/dL(6.7 mmol/L)
      - Failed outpatient insulin or oral glucose-lowering medication management ≥ 5d [19]
      - Blood glucose monitoring
      - Insulin or oral glucose-lowering medication initiation or adjustment [20, 21, 22, 23]
    - ⊖ Vaginal bleeding ≤ 1 pad/h without fetal compromise and, **Both:** [24]
      - Hct or Hb monitoring
      - Ultrasound assessment [6]
  - ⊖ **ACUTE, ≥ One:** [25]
    - ⊖ High risk of fetal compromise, **Both:**
      - ⊖ Finding, ≥ **One:**
        - Biophysical profile ≤ 4 [26, 27]
        - Biophysical profile 6 and ≥ 37 wks gestation [26, 28]
        - ⊖ Non-reassuring fetal testing on two or more tests prior to decision to admit and, ≥ **One:** [29]
          - Amniotic fluid index ≤ 5 cm [30]
          - Biophysical profile 6 and < 37 wks gestation [26, 31]
          - Nonreactive nonstress test [32]
          - Single deepest vertical pocket of amniotic fluid ≤ 2 cm [33]
          - Positive contraction stress test [34]
        - ⊖ Risk factor, ≥ **One:**
          - Chorioamniotic separation (CAS) [35]
          - ⊖ Fetal growth restriction and, ≥ **One:** [36]
            - Absent end-diastolic flow [37]
            - Reverse end-diastolic flow [38]
            - Oligohydramnios [39]
          - Monochorionic-monoamniotic twins and expectant management [40]
          - Multiple gestation and complication [41, 42, 43]
          - Prolapse of amniotic sac [44]
          - ⊖ Fetal anemia and, ≥ **One:** [45]
            - Complication of intrauterine transfusion
            - Fetal hydrops
      - ⊖ Intervention, ≥ **One:**
        - Fetal blood product transfusion
        - ⊖ Monitoring, **All:**

- Fetal heart rate monitoring [5]
- Ultrasound assessment [6]
- Uterine monitoring [9]

- Placental complications and, **Both:** [46, 47, 48, 49, 50]
  - Finding, **One:**
    - Bleeding, contractions, or pain
    - Asymptomatic and expectant management, ≥ **One:** [51]
      - Limited access to prompt medical care and arrangements for outpatient services pending ≤ 24h [52]
      - Outpatient services unavailable
      - Patient or caregiver unreliable and unable to adhere to outpatient instructions or treatment [53]
      - Severe and persistent mental illness, cognitive impairment, or substance use disorder [54]
  - Intervention, ≥ **One:**
    - Fetal heart rate monitoring or uterine monitoring [5, 9]
    - IV fluid or blood product transfusion [13]
- Preterm labor, **All:** [8, 50, 55]
  - Gestation < 37 wks
  - Contractions ≤ every 10 min for ≥ 30 sec
  - Cervical dilation or effacement
  - Fetal heart rate monitoring [5]
  - Uterine monitoring [9]
- Preterm prelabor rupture of membranes (PPROM) confirmed and, **All:** [56, 57, 58, 59, 60]
  - Gestation < 37 wks
  - Fetal heart rate monitoring [5]
  - Uterine monitoring [9]
- Vaginal bleeding > 1 pad/h and, **All:**
  - Hct or Hb monitoring
  - IV fluid or blood product transfusion [13]
  - Ultrasound assessment [6]

- **CRITICAL, ≥ One:** [61]
  - For hypertension (**use Hypertensive Disorders of Pregnancy subset**)
  - For seizures related to eclampsia (**use Hypertensive Disorders of Pregnancy subset**)
  - DIC and blood product transfusion [62]
  - ECMO or ECLS [63]
  - Hemodynamic instability, **All:** [64]
    - ED treatment, ≥ **One:**
      - ≥ 2 units blood product transfusion
      - ≥ 2 liters IV fluid [65]
      - ≥ 1 unit blood product transfusion and ≥ 1 liter IV fluid
    - Findings after ED treatment, ≥ **One:**
      - Heart rate > 120/min
      - Systolic blood pressure < 100 mmHg and < baseline
      - MAP < 65 mmHg
    - Requiring ongoing resuscitation
  - Hemodynamic monitoring, invasive [66]
  - HFNC [67, 68, 69]
  - Mechanical ventilation
  - NIPPV [70]
  - Oxygen ≥ 40%(0.40) [67]
  - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

- **Episode Day 2, One:** [2, 71]
  - **OBSERVATION, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** [72]
      - Undelivered

- Tolerating PO [73]
- Clinical stability, ≥ **One:**
  - Reassuring fetal testing [74]
  - Blood glucose within acceptable limits [75]
  - Vomiting controlled
  - Post cerclage and Category I fetal heart rate tracing [76]
  - Labor ruled out [77]
  - Vaginal bleeding controlled or resolved
- **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
  - Persistent nausea and vomiting of pregnancy or hyperemesis gravidarum and, **Both:** [10, 11]
    - Antiemetic ≥ 2x/24h [12]
    - IV fluid [13]
  - Pregestational or gestational diabetes mellitus and, **All:** [15, 16, 17]
    - Blood glucose monitoring [18]
    - Insulin or oral glucose-lowering medication initiation, adjustment, or dose held [20, 21, 22, 23]
    - Patient or caregiver knowledge deficit and diabetes education [78, 79]
- **ACUTE,** ≥ **One:**
  - Delivery, **One:**
    - **Use Labor and Delivery subset**
    - Planned within 48h and, **Both:** [50]
      - ≥ 22 and < 37 wks gestation
      - Corticosteroid administered within last 48h
  - Fever, **Both:**
    - Temperature ≥ 100.4°F(38.0°C) [80]
    - Intervention, **One:**
      - Known source and anti-infective
      - New onset and, ≥ **One:** [81]
        - Culture pending ≤ 2d
        - Imaging study ≤ 24h
  - High risk of fetal compromise and, ≥ **One:** [82]
    - Fetal blood product transfusion
    - Fetal heart rate or uterine monitoring [5, 9]
  - Placental complications and, ≥ **One:** [46, 47, 48, 49, 50]
    - Fetal heart rate monitoring or uterine monitoring [5, 9]
    - IV fluid or blood product transfusion [13]
  - Preterm labor and fetal heart rate and uterine monitoring [5, 8, 9, 50, 55]
  - Preterm prelabor rupture of membranes (PPROM) requiring fetal heart rate or uterine monitoring [5, 9, 56, 57, 58]
  - Vaginal bleeding and, **Both:**
    - Hct or Hb monitoring
    - IV fluid or blood product transfusion [13]
  - Post Critical care ≤ 24h
- **INTERMEDIATE,** ≥ **One:** [83]
  - HFNC and respiratory monitoring every 3-4h [68, 69]
  - NIPPV [70]
- **CRITICAL,** ≥ **One:**
  - For hypertension (**use Hypertensive Disorders of Pregnancy subset**)
  - For seizures related to eclampsia (**use Hypertensive Disorders of Pregnancy subset**)
  - DIC and blood product transfusion [62]
  - ECMO or ECLS [63]
  - Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [64]
    - Finding, ≥ **One:**

- Heart rate > 120/min
- Systolic blood pressure < 100 mmHg and < baseline
- MAP < 65 mmHg
- ⊖ Intervention, ≥ **One:**
  - Blood product transfusion
  - IV fluid resuscitation ≥ 1L ≤ 24h [65]
- Hemodynamic monitoring, invasive [66, 84]
- HFNC and respiratory monitoring every 1-2h [67, 68, 69]
- Mechanical ventilation
- ⊖ NIPPV and, ≥ **One:** [70]
  - FiO_2 > 50%(0.50)
  - Respiratory intervention every 1-2h [85]
- Oxygen ≥ 40%(0.40) [67]
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

⊖ **Episode Day 3-X, One:** [2, 86]

- ⊖ **OBSERVATION, One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [72]
    - Undelivered
    - Blood glucose within acceptable limits [75]
    - Tolerating PO [73]
  - ⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
    - For hyperemesis gravidarum continued stay use Episode Day 3 at a higher level of care
    - For pregestational or gestational diabetes mellitus continued stay use Episode Day 3 at a higher level of care
    - For a condition not included in the Antepartum subset use appropriate subset (Episode Day 1) based on clinical findings
    - Refer for secondary review
- ⊖ **ACUTE, One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [72]
    - Afebrile
    - Reassuring fetal testing [74]
    - Tolerating PO [73]
    - Undelivered
    - ⊖ Clinical stability, ≥ **One:** [87]
      - Blood glucose within acceptable limits [75]
      - Preterm labor and no cervical dilation or effacement
      - Vaginal bleeding controlled or resolved
      - Vomiting controlled
  - ⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - ⊖ Delivery, **One:**
      - **Use Labor and Delivery subset**
      - ⊖ Planned within 48h and, **Both:** [50]
        - ≥ 22 and < 37 wks gestation
        - Corticosteroid administered within last 48h
    - ⊖ Fever, **Both:**
      - Temperature ≥ 100.4°F(38.0°C) [80]
      - ⊖ Intervention, **One:**
        - Known source and anti-infective
        - ⊖ New onset and, ≥ **One:** [81]
          - Culture pending ≤ 2d
          - Imaging study ≤ 24h
    - ⊖ High risk of fetal compromise and, ≥ **One:** [82]

- Fetal blood product transfusion
- Fetal heart rate monitoring or uterine monitoring [5, 9]
- Hyperemesis gravidarum and, **All:** [11]
  - Finding, ≥ **One:**
    - Inadequate oral intake [88]
    - Vomiting uncontrolled
  - Antiemetic ≥ 2x/24h [12]
  - IV fluid ≤ 4d since initiation [13]
- Placental complications and, **Both:** [46, 47, 48, 49, 50]
  - Finding, ≥ **One:**
    - Bleeding, contractions, or pain
    - Expectant management, ≥ **One:** [51]
      - Limited access to prompt medical care and arrangements for outpatient services pending ≤ 24h [52]
      - Outpatient services unavailable
      - Patient or caregiver unreliable and unable to adhere to outpatient instructions or treatment [53]
      - Severe and persistent mental illness, cognitive impairment, or substance use disorder [54]
  - Intervention, ≥ **One:**
    - Fetal heart rate monitoring or uterine monitoring [5, 9]
    - IV fluid or blood product transfusion [13]
- Pregestational or gestational diabetes mellitus, uncontrolled blood glucose ≤ 24h and, **Both:** [15, 16]
  - Finding, ≥ **One:** [18]
    - Blood glucose < 70 mg/dL(3.9 mmol/L)
    - Blood glucose > 200 mg/dL(11.1 mmol/L)
  - Intervention, **All:**
    - Blood glucose monitoring
    - Insulin or oral glucose-lowering medication initiation, adjustment, or dose held [20, 21, 22, 23]
    - Patient or caregiver knowledge deficit and diabetes education [78, 79]
- Preterm labor, **All:** [8, 50, 55]
  - Gestation < 37 wks
  - Contractions ≤ every 10 min for ≥ 30 sec
  - Cervical dilation or effacement
  - Fetal heart rate monitoring [5]
  - Uterine monitoring [9]
- Preterm prelabor rupture of membranes (PPROM) and, **All:** [56, 58]
  - Gestation < 37 wks
  - Finding, **One:**
    - New onset confirmed [59, 60]
    - Expectant management
  - Intervention, ≥ **One:**
    - Anti-infective (includes PO) [57]
    - Fetal heart rate monitoring or uterine monitoring [5, 9]
- Post Critical care ≤ 24h

Non-responder, not clinically stable for discharge and does not meet partial responder criteria, **One:**
- **Use Extended Stay subset**
- Refer for secondary review

INTERMEDIATE, ≥ One:

Partial responder, not clinically stable for discharge and requires continued stay, ≥ **One:**
- HFNC and respiratory monitoring every 3-4h [68, 69]
- NIPPV [70]

Non-responder, not clinically stable for discharge, **One:**
- Does not meet partial responder criteria or at the Acute level of care, **One:**
  - **Use Extended Stay subset**

- Refer for secondary review

- **CRITICAL, One:**
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - For hypertension (**use Hypertensive Disorders of Pregnancy subset**)
    - For seizures related to eclampsia (**use Hypertensive Disorders of Pregnancy subset**)
    - DIC and blood product transfusion [62]
    - ECMO or ECLS [63]
    - Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [64]
      - Finding, ≥ **One:**
        - Heart rate > 120/min
        - Systolic blood pressure < 100 mmHg and < baseline
        - MAP < 65 mmHg
      - Intervention, ≥ **One:**
        - Blood product transfusion
        - IV fluid resuscitation ≥ 1L ≤ 24h [65]
    - Hemodynamic monitoring, invasive [66, 84]
    - HFNC and respiratory monitoring every 1-2h [67, 68, 69]
    - Mechanical ventilation
    - NIPPV and, ≥ **One:** [70]
      - FiO_2 > 50%(0.50)
      - Respiratory intervention every 1-2h [85]
    - Oxygen ≥ 40%(0.40) [67]
    - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
  - **Non-responder**, not clinically stable for discharge, **One:**
    - Does not meet partial responder criteria or at the Acute or Intermediate level of care, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review

Discharge, Level of Care, One: [89]

- **HOME, Both:**
  - Level of care appropriateness, **Both:**
    - Home environment safe and accessible [90]
    - Patient or caregiver demonstrates ability to manage care
  - Discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [91]
    - Medication reconciliation [92]
    - Follow-up care planned [93]
    - Identify and address transportation needs
- **HOME CARE, Both:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible [90]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [94, 95]
    - Skilled services, ≥ **One:** [96]
      - Skilled nursing [97]
      - Skilled therapy, PT, OT, or ST
      - Behavioral health [98]
  - Discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [91]
    - Medication reconciliation [92]
    - Follow-up care planned [93]
    - Identify and address transportation needs

- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [99]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [100]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [101]
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 1-2x/24h [97]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [102, 103]
        - Functional impairments requiring at least supervision [104]
        - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [92]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [97]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [102, 103]
        - ≥ 2 functional impairments requiring at least minimum assistance [104]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [92]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [102, 103]
      - ≥ 2 functional impairments requiring at least minimum assistance [104]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [92]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk

- Services, **All:**
  - Goal directed therapy with rehabilitation potential [102, 103]
  - ≥ 2 functional impairments requiring at least minimum assistance [104]
  - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [105]
- Complete prior to facility transfer, **All:**
  - Medication reconciliation [92]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

- **LONG TERM ACUTE CARE, Both:**
  - Level of care appropriateness, **Both:**
    - Treatment precluded in a lower level
    - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
      - Primary condition or illness and treatment of active comorbid condition(s) [106]
      - Ventilator dependent ≥ 6h/d and weaning planned [107, 108]
      - Acute and comorbid conditions requiring prolonged hospitalization [109]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [92]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
817 OTHER ANTEPARTUM DIAGNOSES WITH O.R. PROCEDURES WITH MCC	**n/a**	**4.3**	**CMS GMLOS**
818 OTHER ANTEPARTUM DIAGNOSES WITH O.R. PROCEDURES WITH CC	**n/a**	**2.4**	**CMS GMLOS**
819 OTHER ANTEPARTUM DIAGNOSES WITH O.R. PROCEDURES WITHOUT CC/MCC	**n/a**	**1.9**	**CMS GMLOS**
831 OTHER ANTEPARTUM DIAGNOSES WITHOUT O.R. PROCEDURES WITH MCC	**n/a**	**3.3**	**CMS GMLOS**
832 OTHER ANTEPARTUM DIAGNOSES WITHOUT O.R. PROCEDURES WITH CC	**n/a**	**2.5**	**CMS GMLOS**
833 OTHER ANTEPARTUM DIAGNOSES WITHOUT O.R. PROCEDURES WITHOUT CC/MCC	**n/a**	**1.9**	**CMS GMLOS**
GESTATIONAL DIABETES MELLITUS	**53%**	**2.1**	**InterQual**
HYPEREMESIS GRAVIDARUM	**56%**	**2.5**	**InterQual**
PLACENTA ABRUPTIO	**5%**	**2.6**	**InterQual**
PLACENTA PREVIA	**13%**	**2.9**	**InterQual**
PRETERM LABOR	**56%**	**1.7**	**InterQual**
PRETERM PREMATURE RUPTURE OF MEMBRANES (PPROM)	**1%**	**3**	**InterQual**

Condition or Procedure	% Pd Obs	LOS (days)	Type
VAGINAL BLEEDING	**65%**	**1.9**	**InterQual**

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:

1:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.

- **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Care Facilitation:

- Ensure the patient's symptoms, concerns, and needs are heard and include the patient as part of the care team (Behm et al., J Womens Health (Larchmt) 2022, 31: 1677-85)
- Complete cardiac, hypertension, preeclampsia, obstetric hemorrhage, sepsis/shock, and substance abuse risk assessment and manage patient according to protocols (Chambers et al., J Cardiovasc Dev Dis 2022, 9:; McCutcheson-Adams, Cardiac Conditions in Obstetric Care Change Package. 2022)
- Ensure patient receives respectful and equitable care (Sufrin, ACOG Committee Opinion: Importance of Social Determinants of Health and Cultural Awareness in the Delivery of Reproductive Health. Reaffirmed 2021)
    - Provide access to interpreter services
    - Assess patient's health literacy
- Educate patients on urgent warning signs and access to immediate medical care when applicable,

including (Behm et al., J Womens Health (Larchmt) 2022, 31: 1677-85; Friedman et al., AJP Rep 2018, 8: e79-e84):
  - Headache that will not go away or worsens over time
  - Dizziness or fainting
  - Thoughts about hurting yourself or your baby
  - Changes in vision
  - Fever
  - Trouble breathing
  - Chest pain or fast heart rate
  - Severe and persistent belly pain
  - Severe nausea and vomiting
  - Baby's movements stop or slow down
  - Vaginal bleeding or fluid leaking
  - Swelling, redness, or pain in leg(s)
  - Extreme swelling of hands or face
  - Overwhelming tiredness

- Assess patient for chronic health conditions (e.g., hypertension, diabetes, depression, congenital heart disease, cardiovascular risk) and manage appropriately (Chambers et al., J Cardiovasc Dev Dis 2022, 9:; McCutcheson-Adams, Cardiac Conditions in Obstetric Care Change Package. 2022)
- Screen all pregnant and postpartum people for substance abuse disorders and refer to community services and resources
- Discharge with postpartum care plan and use communication hand-off tools including (Obstet Gynecol 2018, 131: e140-e50):
  - Contact information for care team
  - Follow-up appointment time, date, location, and phone number (best practice is phone or telehealth appointment 3 weeks postpartum with comprehensive visit 6-12 weeks postpartum)
  - Infant feeding plan with contact to community support (e.g., Women, Infants and Children (WIC), lactation services, mother support groups)
  - Reproductive plan (e.g., access to contraception)
  - Education on potential complications and postpartum problems
  - Treatment plan for chronic health conditions and contact information for specialist care team
  - Anticipatory guidance on signs and symptoms of perinatal depression or anxiety and management of previous mental health issues
  - Referral to specialist care, mental health support, and substance use disorder treatment if needed

- Screen for SDOH and refer to social services and/or community resources (Billoux, NAM Perspectives 2017:)
  - **Social Determinants of Health (SDOH)** are conditions in a person's environment that may affect their health. Social needs related to SDOH may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social needs should be assessed on admission to determine which interventions are relevant to the individual and their family support system. This list may not be all inclusive, and all relevant SDOH factors should be considered. Any social need determined to be relevant to a patient and family should be planned for throughout the hospital stay. The following interventions should be considered for all patients and families if social need is relevant.
  - **Alcohol or substance use:**
    - Coordinate care with outpatient primary care or behavioral health providers
    - Refer to addiction social support systems
    - Provide information about community resources and programs for family-centered treatment (e.g., intensive case management, detox, outpatient, residential treatment, day treatment, and medication-assisted treatment)
    - Provide information about home visitation programs (e.g., Parent-Child Assistance Program (PCAP) for pregnant and parenting individuals with substance use disorders after care) (Hildebrandt et al., Infant Ment Health J 2020, 41: 677-96; Vanderplasschen et al., Front Psychiatry 2019, 10: 186)
    - Provide information about cognitive behavioral therapy (CBT), multidimensional family therapy (MDFT), and/or pharmacotherapy for youth with co-occurring

emotional/mental disorders (Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021)

- **Child maltreatment, abuse, or intimate partner abuse, sexual, and/or psychological abuse:**
    - Arrange for alternative housing situation, foster care, or shelter if home environment is unsafe
    - Assist with creating a long-term safety plan or make appropriate referral
    - Refer to child protective services (CPS) and other domestic violence services if warranted
    - Refer for psychosocial support if warranted
    - Provide community resources and legal information
    - Provide information on parent training programs such as the Nurse-Family Partnership (NFP) (Nurse-Family Partnership National Resources for Families. 2021; Gubbels et al., Int J Environ Res Public Health 2019, 16:)

- **End of life planning:**
    - Arrange palliative care or hospice
    - Consider palliative care for children with end-stage disease
    - Consider hospice care for terminally ill children with a life expectancy of less than 6 months

- **Environmental exposure to toxic substances and other physical hazards:**
    - Lead poisoning: Provide community resources for primary prevention and secondary prevention strategies (e.g., blood lead level testing, follow-up) (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)
    - Tobacco use (e.g., cigarettes, e-cigarettes, vaping): Provide education and/or brief counseling about tobacco use cessation and the risk of second-hand smoke (U.S. Department of Health and Human Services, Tobacco Use in Children and Adolescents: Primary Care Interventions. 2021; Opinion, Tobacco and Nicotine Cessation During Pregnancy. 2017)

- **Financial resources strained or limited:**
    - *Food insecurity*
        - Provide information about local food banks and special nutritional programs (e.g., Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), U.S. Department of Agriculture Summer Food Service Program) (Testa and Jackson, Ann Epidemiol 2021, 58: 22-8; Opinion, Tobacco and Nicotine Cessation During Pregnancy. 2017)
        - Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)

    - *Income insecurity*
        - Provide information about local food banks and special nutritional programs (e.g., Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), U.S. Department of Agriculture Summer Food Service Program) (Testa and Jackson, Ann Epidemiol 2021, 58: 22-8; Opinion, Tobacco and Nicotine Cessation During Pregnancy. 2017)
        - Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)

- **Infants and children with special health care needs:**
    - Provide anticipatory guidance and education to caregivers regarding medical condition and expected progress after discharge over the next few days, months, or years, as warranted (Dosman and Andrews, Paediatr Child Health 2012, 17: 75-80)
    - Refer for follow-up care to neonatologist or other medical subspecialist for infants

or children with ongoing and/or chronic medical issues (e.g., bronchopulmonary dysplasia, feeding dysfunction) (Pediatrics 2008; 122(5): 1119-1126)
- Refer for adaptive/assistive medical equipment and equipment for technology-dependent children (e.g., home oxygen, ventilators, monitors, tube feeding) (Pediatrics 2008; 122(5): 1119-1126)
- Refer for neurodevelopmental assessment for high-risk infants
- Refer to post-discharge follow-up programs (e.g., Maternal, Infant, and Early Childhood Home Visiting Program, Early Head Start program for infants or toddlers under the age of 3 and pregnant individuals in need of medical, mental health, nutrition, and education, or the Nurse-Family Partnership for mothers pregnant with their first child and for ongoing nurse care visits through the child's second birthday) (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-Family Partnership National Resources for Families. 2021)
- Refer for behavioral support and/or counseling for caregivers through therapy groups, telephone support, and/or peer support groups for caregivers of children with similar health needs (Reyes et al., Health Equity 2021, 5: 100-18; Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021; Pierron et al., BMC Public Health 2018, 18: 1087)
- Refer to lactation consultant in the hospital and home visits for infants at risk for poor lactation or suboptimal infant breast-feeding behavior (Leeman et al., Pediatr Qual Saf 2019, 4: e130)
- Refer for respite services, if warranted

- **Mental health comorbidities**:
    - Arrange appointment with primary care provider or behavioral health specialist upon discharge
    - Coordinate care with outpatient behavioral health providers
    - Provide information and resources about suicide prevention, including the 988 Suicide and Crisis Lifeline
    - Refer for psychiatric services for suicide risk screening and intervention
    - Refer to school-based mental health interventions, if applicable (Fazel et al., Lancet Psychiatry 2014, 1: 377-87)
    - Provide information about intensive community-based treatment (e.g., functional family therapy, intensive child and adolescent and psychiatric services, multidimensional family therapy, intensive case management)
    - Transfer to day treatment program, residential addiction services, or partial hospitalization when warranted

- **Risk for treatment nonadherence**:
    - Provide information about educational and behavioral support initiatives for children/individuals at risk of medication nonadherence and their caregivers, such as text message reminders to take medication and electronic medication dispensers
    - Provide information about peer support groups for children, individuals, and/or families with poor insight into illness or noncompliance with treatment (Mental Health America, Peer Support: Research and Reports 2021; National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
    - Refer to interventions to prevent medical relapse or readmission, and to enhance medication compliance

- **Risk for readmission:**
    - Consider transfer to an alternative level of care such as subacute care
    - Consider home care services
    - Consider early childhood home visiting programs (e.g., Early Head Start; Early On; Nurse-Family Partnership; and the Maternal, Infant, and Early Childhood Home Visiting Program) (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-

Family Partnership National Resources for Families. 2021)
  - Consider telehealth and web-based training programs (e.g., telepsychiatry or telebehavioral health and mobile health interventions) (American Academy of Child and Adolescent Psychiatry, J Am Acad Child Adolesc Psychiatry 2017, 56: 875-93)
  - Consider intensive in-home behavioral health services
  - Provide patient and family with condition-specific information, including appropriate planning for prevention of respiratory syncytial virus (RSV) infection and appropriate immunizations

- **Special health risk behaviors in adolescence and young adulthood**:
  - Arrange follow-up appointment with primary care provider and/or specialists upon discharge
  - Coordinate care with outpatient behavioral health providers
  - Provide information on mentoring programs (e.g., Big Brothers Big Sisters of America)
  - Refer for intensive community-based treatment for individuals with comorbid behavioral health issues
  - Refer to community resources for pregnancy prevention/family planning (e.g., use of birth control, subsequent pregnancies) (Maness et al., J Adolesc Health 2016, 58: 636-43)

- **Eating disorders:**
  - Provide information on nutrition services, including education and resources for diet management, physical activity, and promotion of healthy eating attitudes and behaviors (Gato-Moreno et al., Int J Environ Res Public Health 2021, 18:; Al-Khudairy et al., Cochrane Database Syst Rev 2017, 6: Cd012691)
  - Provide information on outpatient psychosocial interventions and family-based treatment (Lock and La Via, J Am Acad Child Adolesc Psychiatry 2015, 54: 412-25)

- **Parenting resources:**
  - Provide information on family resource centers and parenting support groups for young, first-time parents (e.g., National Parent Helpline) (National Parent and Youth Helpline)
  - Provide information on community resources to improve parenting skills (U.S. Department of Health and Human Services, Parenting Resources. 2021)

- **Unstable living environment or homelessness expected upon discharge:**
  - Assist with health insurance application
  - Provide information about homeless shelter application if applicable
  - Refer to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
  - Refer to federally sponsored outpatient programs
  - Refer to intensive case management intervention, assertive community treatment, or other intensive community treatment (Ponka et al., PLoS One 2020, 15: e0230896; Vijverberg et al., BMC Psychiatry 2017, 17: 284)

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

Reduced or absent fetal movement is associated with adverse fetal outcomes and benefits from fetal surveillance, assessment, and monitoring to detect potential uteroplacental compromise and fetal decompensation to allow for the initiation of timely interventions (Obstet Gynecol 2021, 137: e116-e27; Dore and Ehman, J Obstet Gynaecol Can 2020, 42: 316-48 e9).

5:

The fetal heart rate (FHR) fluctuates constantly in response to changes in the intrauterine environment and other stimuli (e.g., uterine contractions). Surveillance of FHR and FHR patterns is used to assess fetal well-being and oxygenation to identify potential uteroplacental and fetal compromise, allowing for intervention, prevention, and mitigation of adverse outcomes (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27; Devane et al., Cochrane Database Syst Rev 2017, 1: CD005122). Standard FHR monitoring techniques include auscultation, Doppler ultrasound, and electronic fetal monitoring that captures the FHR on a graphical printout called a cardiotocograph using an ultrasound transducer externally attached to the maternal abdomen or an electrode placed directly on the fetal scalp internally (Devane et al., Cochrane Database Syst Rev 2017, 1: CD005122). FHR monitoring is a component of the nonstress test, contraction stress test, biophysical profile, and modified biophysical profile (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

6:

Ultrasound is utilized to evaluate fetal well-being, anatomy, growth, position, movement, tone, breathing movements, amniotic fluid volume, chorionicity, placentation, and umbilical artery flow (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27; American College of and Gynecologists' Committee on Practice Bulletins-Obstetrics, Obstet Gynecol 2021, 137: e145-e62; Obstet Gynecol 2020, 135: e237-e60).

7:

Labor is classified into stages (first, second, and third) and phases (latent and active). The first stage of labor is comprised of the latent and active phases. The maternal perception of regular, consistent contractions marks the beginning of the latent phase of labor. The active phase of labor is thought to commence when the rate of cervical dilation increases significantly and the cervix is at least 6 cm dilated (American College of et al., Am J Obstet Gynecol 2014, 210: 179-93). Admission to labor and delivery may be deferred for pregnant individuals in the latent phase of labor who are stable with reassuring fetal status (Obstet Gynecol 2019, 133: e164-e73)

8:

The phrase "preterm birth" refers to the act of giving birth at or after 20 0/7 weeks of gestation but before 37 0/7 weeks of gestation. The occurrence of preterm birth can be attributed to various factors, including spontaneous events such as preterm labor, premature prelabor rupture of membranes, or cervical insufficiency. Additionally, preterm birth may be indicated as a result of the presence of certain maternal or fetal conditions or complications (Obstet Gynecol 2021, 138: e65-e90). The leading cause of hospitalization and infant mortality is preterm birth. The presence of regular uterine contractions and a change in cervical dilation, effacement, or both, or the presence of regular contractions with cervical dilation of at least 2 centimeters on initial presentation, are diagnostic of preterm labor (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2016, 128: e155-64). Research has demonstrated that cervical shortening identified on ultrasonographic assessment is associated with an increased risk of preterm birth (Obstet Gynecol 2021, 138: e65-e90; Obstet Gynecol 2014, 123: 372-9).

9:

Tocography is used to detect uterine contractions and is most commonly performed externally, although internal monitoring may be indicated in select cases. External tocography, which records uterine activity, involves the placement of a transducer over the uterine fundus.

10:

The prevalence rates for nausea and vomiting during pregnancy are 50-80%(0.50-0.80) for nausea and 50%(0.50) for vomiting and retching. Early treatment of pregnancy-related nausea and vomiting can prevent the development of hyperemesis gravidarum. Hyperemesis gravidarum is a diagnosis of exclusion, with the most frequently reported findings being persistent vomiting unrelated to other causes, evidence of acute malnutrition (e.g., large ketonuria), and evidence of weight loss (e.g., at least 5%(0.05) of pre-pregnancy weight). Additionally, electrolyte, thyroid, and liver abnormalities may exist. One of the most common reasons for hospitalization during pregnancy is hyperemesis gravidarum (Committee on Practice, Obstet Gynecol 2018, 131: e15-e30).

11:

Hospitalization for evaluation and management of dehydration and electrolyte imbalance is advised when a pregnant individual cannot tolerate oral intake without vomiting and has not responded to outpatient treatment (Committee on Practice, Obstet Gynecol 2018, 131: e15-e30).

12:

Pregnant individuals with nausea and vomiting of pregnancy who cannot tolerate oral intake and medications may benefit from other medication administration routes, such as intravenous, intramuscular, transdermal, or rectal suppositories. Pharmacotherapy for persistent symptoms may include (Committee on Practice, Obstet Gynecol 2018, 131: e15-e30):

- Dopamine antagonists (e.g., metoclopramide, promethazine, prochlorperazine, chlorpromazine)
- Antihistamines (e.g., dimenhydrinate, diphenhydramine)
- Serotonin antagonists (e.g., ondansetron)
- Steroids (e.g., methylprednisolone)

Second-line pharmacotherapy may include rectally or orally administered dopamine antagonists or oral antihistamines when symptoms persist. If dehydration ensues, intravenous fluid replacement and intravenous pharmacotherapy is recommended (Committee on Practice, Obstet Gynecol 2018, 131: e15-e30).

13:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

14:

Cervical insufficiency refers to the uterine cervix's failure to maintain a pregnancy after 20 weeks of gestation in the absence of contractions and/or labor. Transvaginal and transabdominal cervical cerclage are surgical treatment options for cervical insufficiency, recommended in the second trimester before fetal viability. Transabdominal cerclage increases the risk of potentially life-threatening hemorrhage and the standard complications associated with abdominal surgery. Additional complications for cerclage may include rupture of membranes, chorioamnionitis, uterine rupture, and septicemia (Obstet Gynecol 2014, 123: 372-9). Hospitalization for post-procedure surveillance may be necessary to monitor for complications.

15:

Pregestational diabetes mellitus (DM) is either type 1 or 2 DM that is present before pregnancy, also known as preexisting DM (Ornoy et al., Int J Mol Sci 2021, 22:; Rosene-Montella, In: Goldman and Schafer, ed. Goldman-Cecil Medicine, 26th edn. 2020:1584-5).

16:

Gestational diabetes mellitus (DM) refers to DM that is not present before gestation, and diagnosis occurs in the second or third trimester of pregnancy (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

17:

InterQual® external peer reviewers agree that gestational and pregestational (preexisting) diabetes mellitus (DM) can be managed in the outpatient setting with medication, diet modification, and exercise. However, hospitalization may be appropriate for pregnant individuals with persistent uncontrolled blood glucose despite treatment. Pregnant individuals with DM have an increased risk for fetal, neonatal, or perinatal mortality, macrosomia, preterm labor, congenital malformations, and the risk of respiratory distress syndrome in the newborn (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; ACOG. Committee on Practice, Obstet Gynecol 2017, 130(1): e17-e31).

18:

Recommendations are that blood glucose monitoring occurs while fasting, postprandially, and in some cases preprandially, for improved glycemic control in pregnancy. According to the American Diabetes Association, target blood glucose levels for pregnant individuals with type 1 or type 2 diabetes mellitus (DM) are a fasting plasma glucose less than 95 mg/dL(5.3 mmol/L) and a 1-hour postprandial glucose less than 140 mg/dL(7.58 mmol/L) or a 2-hour postprandial glucose less than 120 mg/dL(6.7 mmol/L). Lower limits do not apply to pregnant individuals with type 2 DM controlled with diet. Glycemic targets are stricter in pregnancy; therefore, pregnant individuals with DM must have adequate and consistent carbohydrate intake that aligns with insulin dosing to prevent hyperglycemia or hypoglycemia. If these targets cannot be achieved safely without significant hypoglycemia, recommendations are to use clinical judgment and expertise to individualize care and targets. Per the American Diabetes Association 2023 guidelines, an individualized target A1C in pregnancy should be less than 0.7%(0.007) if there is a risk for significant hypoglycemia and less than 0.6%(0.006) if there is no risk for significant hypoglycemia. Increased red blood cell turnover in pregnancy can cause A1C levels to fall. Macrosomia risk increases with postprandial hyperglycemia, which A1C does not adequately capture. Therefore, A1C is a secondary indicator of glycemic control in pregnancy. Risks that individuals with pregestational (preexisting) DM can experience include, but are not limited to, spontaneous abortion, macrosomia, fetal anomalies and demise, neonatal respiratory distress syndrome and hypoglycemia, hyperbilirubinemia, and preeclampsia (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

19:

Failed outpatient management is considered when the patient has not responded to treatment received through the medical practitioner's office, outpatient clinic, or a trial of home care services and includes care delivered in the Observation setting.

20:

Instruction: Medication adjustment includes initiation of a new drug, change in dosage, and/or administration frequency of a medication. Blood glucose values guide medication dosing decisions and adjustments. Administration of sliding scale insulin must occur at least 3 times within 24 hours. A continuous infusion rate must have at least 2 rate adjustments within 24 hours (including insulin pump rate adjustments). A glucose-lowering medication dose withheld based on normal or low blood glucose levels is included in the count of doses administered within the 24-hour timeframe.

21:

Pregnant individuals with pregestational (preexisting) type 1 diabetes mellitus (DM) are usually treated with insulin. Increased insulin sensitivity is present during early pregnancy and insulin resistance increases around weeks 16-36 of pregnancy. When there is increased insulin sensitivity, pregnant individuals with type 1 DM will have an increased risk for hypoglycemia and may require lower insulin dosages. Alternatively, insulin dosage requirements may increase in pregnancy when insulin resistance increases. Insulin titration and frequent daily blood glucose monitoring are necessary due to the physiological insulin sensitivity changes occurring during pregnancy (American Diabetes Association, Diabetes Care 2024, 47: S1-S314). In pregnant individuals with pregestational type 2 DM or gestational DM, insulin may be needed for adequate glycemic control if blood glucose monitoring, physical activity, weight management, and medical nutrition therapy are not sufficiently achieving blood glucose targets. For pregnant individuals with pregestational or gestational DM who require medical therapy, insulin is the preferred agent. However, it is essential to note that insulin may not be feasible if these patients have specific religious or cultural beliefs or financial, psychosocial, or literacy barriers. Oral glucose-lowering medication may be an alternative for these patients after discussing the risks involved, as these medications can cross the placenta (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2018, 132: e228-e48; Obstet Gynecol 2018, 131: e49-e64).

22:

Scheduled subcutaneous insulin is recommended to manage hyperglycemia in non-critically ill hospitalized patients. The preferred treatment for patients with adequate nutritional intake includes an insulin regimen that consists of basal, prandial, and correction insulin aligned with meals. Recommendations are to complete point-of-care (POC) glucose testing and subcutaneous rapid or short-acting insulin administration immediately before meals. Alternatively, in patients with inadequate oral intake, prandial insulin administration is recommended directly after the patient eats, with adjustments in dosing related to the amount of food consumed. The preferred treatment for patients with oral intake restrictions includes basal insulin or basal plus correction insulin. For these patients or those on continuous enteral or parenteral nutrition, recommendations are to complete POC glucose testing and subcutaneous rapid or short-acting insulin administration every 4-6 hours (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

23:
When making treatment decisions, providers should assess for socioeconomic and psychosocial barriers and consider medication-taking behavior and the religious and cultural preferences of the patient. For non-critically ill patients with hyperglycemia and type 2 diabetes mellitus (DM) in the hospital, insulin is often the preferred treatment. However, based on patient preference, it can be appropriate to give oral glucose-lowering medication. If deemed appropriate by the provider, patients with preexisting DM may continue their home regimen, including oral glucose-lowering medication. When oral glucose-lowering medication is on hold during hospitalization, resumption of this medication may be suggested 1-2 days prior to discharge (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

24:
The term "fetal compromise" refers to metabolic abnormalities (e.g., hypoxemia, acidosis) that can lead to unfavorable consequences for the developing fetus (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

25:
Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

26:
The biophysical profile (BPP) involves evaluating a nonstress test, fetal respiration movements, fetal movement, fetal tone, and amniotic fluid volume. Each element receives a score of 2 (present) or 0 (absent). A total combined score of 8 or 10 is normal, a score of 6 is equivocal, and a score of 4 or less is abnormal (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

27:
In most cases, a biophysical profile score of 4 necessitates delivery; however, in pregnancies less than 32 weeks gestation, care should be individualized and extended monitoring may be reasonable. If delivery is not planned due to an early gestational age, the results of additional antenatal surveillance will not inform care (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

28:
A biophysical profile score of 6 in a fetus at 37 weeks of gestation or beyond usually warrants additional assessment and consideration for delivery (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

29:
Antepartum fetal surveillance is used to assess the risk of fetal death in pregnancies at high risk for stillbirth or with multiple comorbidities. Methods of detecting fetal compromise include assessing fetal movement, fetal heart rate monitoring, biophysical profile, nonstress testing, contraction stress testing, and umbilical artery Doppler velocimetry. Fetal surveillance tests have high false-positive rates and low positive predictive value; therefore, abnormal test results should be followed by another test to confirm findings. Antenatal fetal testing used to

determine the decision to admit should have been performed no more than one week prior to decision to admit (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27; American College of and Gynecologists' Committee on Obstetric Practice, Obstet Gynecol 2021, 137: e177-e97).

30:

The volume of amniotic fluid is calculated using ultrasound. Oligohydramnios is often defined as having an amniotic fluid index of 5 cm or less and a single deepest vertical pocket of amniotic fluid of 2 cm or less. According to the available evidence from randomized controlled trials, using the single deepest vertical pocket measurement for detecting oligohydramnios instead of the amniotic fluid index is linked to fewer unnecessary interventions without an associated rise in adverse perinatal outcomes. For pregnant individuals with oligohydramnios the decision to proceed with expectant management or delivery should be individualized based on gestational age and the maternal and fetal condition. If the plan is to not proceed with delivery, follow-up amniotic fluid volume measurements, nonstress tests, and fetal growth assessments are indicated (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

31:

A repeat biophysical profile (BPP) should be obtained in 24 hours for a fetus that is less than 37 weeks gestation with a BPP score of 6 (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

32:

The nonstress test (NST) is centered on the principle that the heart rate of a fetus that is neither acidotic nor neurologically depressed will increase briefly in conjunction with fetal movement. The results of the NST are classified as reactive or nonreactive. A reactive NST consists of two or more fetal heart rate (FHR) accelerations occurring within twenty minutes. A nonreactive NST is one with insufficient FHR accelerations over the course of forty minutes. Decelerations in FHR during an NST that continue for 1 minute or longer are associated with a significantly increased risk of cesarean delivery and fetal mortality (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

33:

The volume of amniotic fluid is calculated using ultrasound. A single deepest vertical pocket of amniotic fluid that is 2 cm or less and does not contain the umbilical cord or fetal extremities is considered diagnostic of oligohydramnios, which should necessitate further evaluation. For pregnant individuals with oligohydramnios, the decision to proceed with expectant management or delivery should be individualized based on gestational age and the maternal and fetal condition. If the plan is to not proceed with delivery, follow-up amniotic fluid volume measurements, nonstress tests, and fetal growth assessments are indicated (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

34:

The contraction stress test (CST), also known as the oxytocin challenge test or stress test, measures the fetal heart rate (FHR) in response to uterine contractions. The CST is predicated on the understanding that fetal oxygenation will temporarily decrease during uterine contractions and that inadequate fetal oxygenation will result in late FHR decelerations. If spontaneous uterine contractions are not present, they are induced via nipple stimulation or intravenous oxytocin administration, while FHR and uterine contractions are monitored. Results of the CST are evaluated according to the presence or absence of late FHR decelerations and classified as negative, positive, equivocal-suspicious, equivocal, and unsatisfactory. A positive CST result indicates that at least half of the contractions within a 10-minute period resulted in FHR late decelerations (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

35:

In the first trimester of pregnancy, chorion and amnion separation are expected. Complete fusion of these membranes typically occurs between 14 and 16 weeks gestation (Zhu et al., Prenat Diagn 2020, 40: 1366-74; Govaerts et al., Clin Case Rep 2019, 7: 762-5). Chorioamniotic separation (CAS), also known as chorioamniotic membrane separation, refers to the new separation of the chorion and amnion after the initial fusion beyond 16 weeks gestation. Iatrogenic CAS may occur after invasive procedures such as amniocentesis or fetal surgery and is associated with an increased risk of perinatal outcomes that include spontaneous preterm delivery, amniotic band formation, umbilical cord compromise, and intrauterine fetal demise. Spontaneous CAS may be associated with an increased risk of preterm delivery and adverse perinatal outcomes, including amniotic bands, structural anomalies, aneuploidy, and intrauterine fetal demise. There is no consensus on the optimal follow-up of pregnant individuals with this condition, and inpatient management after viability may be considered for this high-risk population (Zhu et al., Prenat Diagn 2020, 40: 1366-74).

36:

Ultrasonography is used to diagnose fetal growth restriction, which is characterized by an estimated fetal weight or abdominal circumference below the 10^{th} percentile for gestational age. Pregnant individuals with fetal growth restriction require further evaluation, such as amniotic fluid volume assessment and Doppler blood flow studies of the umbilical artery (Obstet Gynecol 2021, 137: e16-e28).

37:

Doppler ultrasonography is a noninvasive method for evaluating the hemodynamic components of vascular resistance in fetal growth restriction-complicated pregnancies. Abnormal Doppler velocimetry testing reflects decreased, absent, or reversed end-diastolic flow (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27). Absent or reversed end-diastolic flow in the umbilical artery is associated with an increased risk of perinatal mortality and can influence decisions regarding the timing of delivery in the setting of fetal growth restriction (Obstet Gynecol 2021, 137: e16-e28). Inpatient care may be considered for a pregnancy complicated by fetal growth restriction and abnormal umbilical artery Doppler waveform characterized by the absence of end-diastolic velocity (American College of and Gynecologists' Committee on Obstetric Practice, Obstet Gynecol 2021, 137: e177-e97; Society for Maternal-Fetal Medicine. Electronic address et al., Am J Obstet Gynecol 2020, 223: B2-B17).

38:

Doppler ultrasonography is a noninvasive method for evaluating the hemodynamic components of vascular resistance in fetal growth restriction-complicated pregnancies. Abnormal Doppler velocimetry testing reflects decreased, absent, or reversed end-diastolic flow (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27). Absent or reversed end-diastolic flow in the umbilical artery is associated with an increased risk of perinatal mortality and can influence decisions regarding the timing of delivery in the setting of fetal growth restriction (Obstet Gynecol 2021, 137: e16-e28). Inpatient care is recommended for pregnant individuals with fetal growth restriction and abnormal umbilical artery Doppler waveforms characterized by reversed end-diastolic flow who are not being delivered (American College of and Gynecologists' Committee on Obstetric Practice, Obstet Gynecol 2021, 137: e177-e97; Society for Maternal-Fetal Medicine. Electronic address et al., Am J Obstet Gynecol 2020, 223: B2-B17).

39:

A single deepest vertical pocket of amniotic fluid that is 2 cm or less and does not contain the umbilical cord or fetal extremities is considered diagnostic of oligohydramnios. An individual whose pregnancy is complicated by fetal growth restriction and a concurrent condition such as oligohydramnios may require inpatient management

(American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

40:

Due to the higher risk of pregnancy complications, women with monochorionic pregnancies are monitored more closely than those with dichorionic pregnancies. Estimates of perinatal mortality in monochorionic-monoamniotic twins range from 12-23%(0.12-0.23). The optimal management of these individuals remains uncertain, and many clinicians offer early inpatient management commencing between 24-28 weeks of gestation (Obstet Gynecol 2021, 137: e145-e62).

41:

- Dichorionic-diamniotic twins
- Higher-order multifetal gestations
- Monochorionic-diamniotic twins
- Monochorionic-monoamniotic twins

42:

Complications may include one or more of the following:
- Fetal growth restriction
- Growth discordance
- Single fetal demise of one twin with surviving twin
- Twin anemia polycythemia sequence (TAPS)
- Twin-to-twin transfusion syndrome (TTTS)
- Twin reversed arterial perfusion (TRAP) sequence
- Umbilical cord entanglement or compression

43:

Multifetal pregnancies increase the risk of maternal, fetal, and neonatal morbidity and mortality. There is a fivefold increase in the risk of stillbirth and a sevenfold increase in the risk of neonatal mortality. Pregnant individuals with multifetal gestations are six times more likely to deliver prematurely and thirteen times more likely to deliver prior to 32 weeks of gestation than those with singleton gestations. Multiple gestation pregnancies are associated with an increased incidence of complications, including but not limited to, fetal and congenital anomalies, prematurity, fetal growth restriction, discordant fetal growth, single fetal demise of one twin, umbilical cord complications, twin-to-twin transfusion syndrome, and twin anemia-polycythemia syndrome (Obstet Gynecol 2021, 137: e145-e62).

44:

Prolapse of the amniotic sac is also known as hourglass membranes, bulging membranes, or prolapsed membranes. Cervical insufficiency, often referred to as cervical incompetence or weakness, is believed to contribute to approximately 15%(0.15) of perinatal losses occurring between the 16th and 26th weeks of gestation. This condition is characterized by the uterine cervix's inability to maintain a pregnancy during the second trimester, without the presence of clinical contractions or labor. Frequently, women exhibit minimal symptoms and develop a spontaneously dilated cervix accompanied by the protrusion of amniotic membranes at or beyond the external os, which causes the membranes to come into close contact with microorganisms present in the vagina. Consequently, there is a heightened susceptibility of infection, which significantly increases the likelihood of experiencing extreme premature birth and subsequent neonatal mortality. The range of therapeutic procedures for managing a pregnancy with cervical insufficiency may include expectant management, termination of pregnancy, or the surgical placement of an emergency cervical cerclage (Hulshoff et al., Am J Obstet Gynecol MFM 2023, 5: 100971).

45:

The most common cause of fetal anemia is maternal alloimmunization and parvovirus B19 infection, although it can be associated with many other conditions. In twin anemia, a chronic intertwin blood transfusion leads to anemia in the donor twin and polycythemia in the recipient. The initial presentation of hydrops secondary to fetal anemia is ascites and later pleural or pericardial effusions, but these are rare (Prefumo et al., Best Pract Res Clin Obstet Gynaecol 2019, 58: 2-14).

46:

For patients in whom pregnancy prolongation is indicated, discharge planning should begin on admission to promote safe transfer to an outpatient setting. Assessment should include the evaluation of patient reliability, logistical barriers, and the feasibility of implementing frequent maternal and fetal surveillance. Obstacles to discharge identified during the assessment require timely action which may include implementing comprehensive patient and/or caregiver teaching, making alternative living arrangements, or establishing follow-up outpatient services.

47:

Placental complications may include one or more of the following:

- Placental abruption
- Placenta accreta spectrum (i.e., placenta accreta, placenta increta, placenta percreta), also known as abnormally invasive placenta
- Placenta previa
- Vasa previa

48:

Placental abruption is the premature separation of the placenta before delivery, which can be a life-threatening obstetric emergency, associated with prematurity, stillbirth, hypoxia, and major congenital anomalies (Riihimaki et al., Pediatrics 2018, 142:). Placental abruption can be classified as either partial or complete (Society for Maternal-Fetal Medicine. Electronic address and Gyamfi-Bannerman, Am J Obstet Gynecol 2018, 218: B2-B8). Although the clinical manifestations may vary, abruption is classically associated with vaginal bleeding and abdominal pain, with or without uterine contractions, and is often accompanied by abnormal fetal heart rate (FHR) patterns. Acute abruption is characterized by the sudden onset of vaginal bleeding, and chronic abruption is characterized by recurrent episodes of light to moderate vaginal bleeding. Acute abruptions that are near-term are generally managed by maternal stabilization followed by delivery. There is a lack of evidence to guide delivery timing and management strategies for chronic abruption. Patients with chronic abruption require close monitoring for recurrent episodes of vaginal bleeding. Initial management focuses on determining the abruption severity and assessing maternal status and stabilization if indicated. In mild and/or chronic abruption, maternal and fetal stability may warrant initial conservative management (Brandt and Ananth, Am J Obstet Gynecol 2023, 228: S1313-S29).

49:

In addition to placental abruption, one or more of the following placental complications may also contribute to bleeding and hemorrhage during pregnancy:

- Placenta accreta refers to abnormal trophoblast invasion of part or all of the placenta into the uterine wall that results in adherence to the myometrium (American College of et al., Obstet Gynecol 2018, 132: e259-e75; Society for Maternal-Fetal Medicine. Electronic address and Gyamfi-Bannerman, Am J Obstet Gynecol 2018, 218: B2-B8)
- Placenta increta refers to the invasion of the placenta into the myometrium (Society for Maternal-Fetal Medicine. Electronic address and Gyamfi-Bannerman, Am J Obstet Gynecol 2018, 218: B2-B8)
- Placenta percreta, also known as abnormally invasive placenta, refers to placental penetration of the

myometrium (Society for Maternal-Fetal Medicine. Electronic address and Gyamfi-Bannerman, Am J Obstet Gynecol 2018, 218: B2-B8)
- Placenta previa refers to the presence of placental tissue directly over the internal cervical os (Jauniaux et al., Bjog 2019, 126: e1-e48; Society for Maternal-Fetal Medicine. Electronic address and Gyamfi-Bannerman, Am J Obstet Gynecol 2018, 218: B2-B8)
- Vasa previa occurs when fetal vessels course through the membranes and traverse the internal os (Fishel Bartal et al., Am J Perinatol 2019, 36: 422-7; Society for Maternal-Fetal Medicine. Electronic address and Gyamfi-Bannerman, Am J Obstet Gynecol 2018, 218: B2-B8)

Placenta accreta spectrum, previously known as morbidly adherent placenta, refers to the various degrees of pathological placental adherence, including placenta increta, placenta percreta, and placenta accreta (American College of et al., Obstet Gynecol 2018, 132: e259-e75). For placental previa, placenta accreta spectrum, and vasa previa, antenatal care, and hospitalization should be determined on an individual basis with consideration of the pregnant individual's needs, resources, and social circumstances due to the increased risk of massive hemorrhage (Jauniaux et al., BJOG 2019, 126: e1-e48).

50:

Antenatal corticosteroid administration is recommended to expedite fetal lung maturation in pregnant individuals at risk for preterm birth to reduce the risk of perinatal death, neonatal death, respiratory distress syndrome, and intraventricular hemorrhage (McGoldrick et al., Cochrane Database Syst Rev 2020, 12: CD004454). The following recommendations are supported by substantial evidence of improved neonatal outcomes:

- For pregnant individuals between 24 weeks and less than 34 weeks gestation who are at risk of preterm delivery within 7 days, a single course of corticosteroids is recommended (Obstet Gynecol 2017, 130: e102-e9)
    - Antenatal corticosteroids may be considered between 22 and 24 weeks of gestation if neonatal resuscitation is planned and appropriate counseling is provided (American Journal of Obstetrics and Gynecology et al., Use of Antenatal Corticosteroids at 22 Weeks of Gestation. 2021; Backes et al., Am J Obstet Gynecol 2021, 224: 158-74; Ehret et al., JAMA Netw Open 2018, 1: e183235)

- For pregnant individuals between 34 and 37 weeks gestation who have not previously received a dose of antenatal corticosteroids, a single course is recommended (Obstet Gynecol 2017, 130: e102-e9)

- For pregnant individuals less than 34 weeks gestation, who are at risk of preterm delivery within 7 days, and who have previously received antenatal corticosteroids more than 14 days ago, a single repeat course of antenatal corticosteroids is recommended (Walters et al., Cochrane Database Syst Rev 2022, 4: CD003935; Obstet Gynecol 2017, 130: e102-e9)

51:

Indications for continued hospitalization may include the following:

- Residence not in close proximity to a hospital
- Lack of transportation or telephone communication
- Inability or unwillingness to follow an outpatient surveillance plan
- Inability to understand the signs of deterioration and recognize the need for prompt medical attention

52:

Limited access to prompt medical care refers to the following:

- Residence not in close proximity to a hospital
- Lack of transportation or telephone communication

53:

Patient unreliable refers to the following:

- Inability or unwillingness to follow an outpatient surveillance plan
- Inability to understand the signs of deterioration and recognize the need for prompt medical attention

54:

The term "severe and persistent mental illness, cognitive impairment" also called "serious and persistent mental illness," refers to a mental illness or cognitive impairment that is characterized by severe symptoms of prolonged duration that require long-term treatment. The illness causes impaired functioning that interferes significantly with primary activities of daily living (ADLs) and results in an inability to maintain independent functioning without treatment, support, and rehabilitation for a prolonged or indefinite period of time. Examples include, but are not limited to, schizophrenia and other psychotic disorders, bipolar disorder, and severe cases of major depression.

55:

Tocolytic therapy may delay delivery long enough to enable the administration of antenatal corticosteroids and magnesium sulfate for neuroprotection and transport to a tertiary care facility, if necessary. Tocolytic therapy is generally efficacious for up to 48 hours; therefore, tocolytic therapy should only be administered to pregnant individuals whose fetuses would benefit from a 48-hour delay in delivery (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2016, 128: e155-64).

56:

Preterm prelabor rupture of membranes (PPROM) refers to membrane rupture before labor that occurs prior to 37 weeks of gestation. Hospitalization is typically advised for pregnant individuals with PPROM in the periviable period (less than 23 to less than 24 weeks gestation), the preterm period (24 to less than 34 weeks gestation), and the late preterm period (34 weeks to less than 37 weeks gestation). When the fetal and maternal status is reassuring, expectant management of pregnant individuals with PPROM may be appropriate, which consists of hospital admission and periodic assessment for infection, placental abruption, umbilical cord compression, fetal well-being, and labor (Obstet Gynecol 2020, 135: e80-e97).
Instruction: Responder criteria are absent for this condition because discharge is rarely indicated due to the high degree of maternal and neonatal risk. In situations where discharge is indicated, the discharge should still occur.

57:

For patients presenting with preterm prelabor or premature rupture of membranes at 24 to 33 completed weeks gestation, guidelines recommend administration of a 7-day course of intravenous ampicillin and erythromycin, followed by oral amoxicillin and erythromycin during expectant management until 34 weeks. The aim of this therapy is to prolong pregnancy and decrease neonatal morbidity. At 34 weeks gestation and beyond, Group B streptococcal (GBS) prophylaxis is recommended. Anti-infective therapy is not routinely recommended for patients less than 24 weeks gestation or GBS negative patients who are 34 weeks gestation or greater (Obstet Gynecol 2019, 134: e19-40; Kuba and Bernstein, Obstet Gynecol 2018, 131: 1163-4).

58:

Preterm prelabor or premature rupture of membranes (PPROM) is defined as the prelabor rupture of membranes prior to 37 weeks gestation (Obstet Gynecol 2020, 135: e80-e97).

59:

Diagnostic testing to confirm rupture of membranes includes evaluation of one or more of the following:

- Fern (+)
- Nitrazine (+)
- Amniotic fluid protein immunoassay (+)
- Pooling of amniotic fluid
- Intra-amniotic dye (+)

60:

The diagnosis of membrane rupture is typically confirmed by a clinical examination involving the observation of amniotic fluid accumulating in the vagina, a vaginal fluid pH test with nitrazine paper, and/or the identification of ferning in dried vaginal fluid upon microscopic examination. Additional tests, such as immunoassays or transabdominal intra-amniotic dye instillation, can aid in the diagnosis of prelabor rupture of membranes and preterm prelabor rupture of membranes in equivocal cases (Obstet Gynecol 2020, 135: e80-e97).

61:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

62:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013:).

63:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

64:

The mainstay of treatment of patients with hemorrhagic shock is blood product transfusion (e.g., packed red blood cells). Despite the general principle of minimizing the use of intravenous fluids in patients with known hemorrhage, crystalloids are often administered until blood products are available, especially in patients who are hypotensive (e.g., mean arterial pressure less than 65). Vital sign changes may underestimate blood loss in patients. Patients with a 30%(0.30) loss of total blood volume may have little to no change in systolic blood pressure (SBP) and only demonstrate modest tachycardia (i.e., heart rate greater than 100). By the time an SBP has dropped to less than 100 mmHg, there may be as much as a 40%(0.40) loss of total blood volume.

65:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).
Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med

2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

66:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

67:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO_2) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO_2 with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO_2 retention, where a precise inspired FiO_2 is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO_2. The following are estimates of O_2 delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(*0.21*)
1L 24%(*0.24*) **4L** 36%(*0.36*)
2L 28%(*0.28*) **5L** 40%(*0.40*)
3L 32%(*0.32*) **6L** 44%(*0.44*)

Simple oxygen masks

- 35-50%(*0.35-0.50*) FiO_2 (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO_2 in the mask

Venturi masks

- 24-50%(*0.24-0.50*) FiO_2 (4-12 L/min)

Partial rebreathing masks

- 40-70%(*0.40-0.70*) FiO_2 (6-10 L/min)

Non-rebreathing masks

- 60-80%(*0.60-0.80*) FiO_2 (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS

Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO_2

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO_2

68:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

69:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

70:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).
The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.
Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

71:

The CMS Two-Midnight Rule generally requires that the medical record support a patient’s need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.
NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

72:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

73:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

74:

Reassuring fetal testing may include any of the following test results that can indicate fetal well-being: reactive nonstress test, biophysical profile (BPP) composite score of 8 or 10, normal modified BPP, single deepest vertical pocket of amniotic fluid volume greater than 2 cm, negative contraction stress test, or high-velocity umbilical artery diastolic flow seen on Doppler velocimetry (American College of and Gynecologists' Committee on Practice, Obstet Gynecol 2021, 137: e116-e27).

75:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

76:

Fetal heart rate (FHR) tracings may be classified into 3 categories (Obstet Gynecol 2009, 114: 192-202):

- Category I (normal): indicative of normal acid-base status at the time of observation
- Category II (indeterminate): no sufficient evidence exists to classify as Category I or Category III, necessitating evaluation, surveillance, and reevaluation
- Category III (abnormal): associated with abnormal fetal acid-base status at the time of observation

77:

Labor can be ruled out when the patient's contractions are controlled and her cervix is unchanged.

78:

National standards recommend diabetes self-management education and support (DSMES) for patients and/or caregivers to enhance knowledge, decision making, and mastery of skills for diabetes management to prevent acute complications, improve treatment outcomes, and reduce risk of long-term complications. Hospitalization represents a critical time to evaluate, assess, and provide DSMES that includes (American Diabetes Association, Diabetes Care 2024, 47: S1-S314):

- Identifying a health care provider to provide diabetes care post discharge
- Establishing home blood glucose goals, self-monitoring of blood glucose, and understanding when to call the provider
- Recognition, treatment, and prevention of hyperglycemia and hypoglycemia
- Administration of blood glucose-lowering medications, including insulin, if relevant
- Information on healthy dietary choices
- Proper use and disposal of insulin needles and syringes
- Sick-day management (e.g., when sick with a cold, flu, or infection)

79:

Critical events such as a new diagnosis, unmet treatment goals, complications, and transitions in life require evaluation of diabetes self-management educational needs to encourage acquiring the skills that may assist in executing and adhering to the treatment regimen and medical nutrition therapy (MNT) plan. An individualized MNT plan created through a joint effort of shared decision making with the patient and the healthcare team that includes a registered dietician nutritionist is essential. The goals of MNT are to attain individualized glycemic targets, delay or prevent complications, reach and maintain body weight goals, incorporate personal and cultural preferences, address barriers to behavioral changes and access to healthy foods, and provide the tools and resources to develop and maintain healthy dietary patterns. Individualized macronutrient recommendations consider the patient's eating patterns, preferences, and metabolic goals, as evidence shows there are no general recommendations for macronutrients for all people with diabetes. Carbohydrate monitoring strategies are essential to achieve glycemic control and include, but are not limited to, carbohydrate counting, an exchange system, or experience-based estimation (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; American Diabetes Association, Diabetes Care 2022, 45:S1-S262).

80:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

81:

A fever is suspected as being new onset when a patient has been afebrile and then develops a new fever or when the fever pattern changes.

82:

High risk for fetal compromise includes one or more of the following:

- Amniotic fluid index ≤ 5
- Biophysical profile ≤ 6
- Chorioamniotic separation
- Fetal anemia
- Fetal hydrops
- Fetal growth restriction with absent end-diastolic flow or reverse-end diastolic flow
- Multiple gestation and fetal growth restriction with growth discordance
- Monochorionic-monoamniotic twins and expectant management
- Nonreactive nonstress test
- Oligohydramnios
- Positive contraction (stress) test
- Prolapse of amniotic sac
- Single deepest vertical pocket ≤ 2
- Single fetal demise of one twin with surviving twin
- Twin anemia-polycythemia sequence (TAPS)
- Twin-twin transfusion syndrome (TTTS)
- Twin reversed arterial perfusion (TRAP) sequence
- Umbilical cord entanglement or compression

83:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

84:

The routine use of invasive hemodynamic monitoring in individuals whose sex at birth is female with severe preeclampsia or eclampsia is not recommended. However, in patients with severe cardiac or renal involvement, hypertension unresponsive to therapy, or pulmonary edema, invasive hemodynamic monitoring may be useful in guiding care (American College of Obstetricians and Gynecologists. Obstet Gynecol 2013, 122: 1122-31).

85:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

86:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or

general subset.
Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status;** those who do not meet criteria would be appropriate for secondary review.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

87:

Patients may be considered for discharge when they have experienced no active bleeding for the last 24h, their cervix is closed or unchanged, their lab values are within acceptable limits, and fetal testing is reassuring. In addition, any treatment needs should be manageable in the post-acute setting; the patient is reliable and medical care is accessible and outpatient services have been arranged.

88:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

89:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

90:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

91:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

92:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record

- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

93:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

94:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

95:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

96:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)

- End-stage disease, hospice, or palliative care

97:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

98:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.
Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

99:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

100:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications

- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

101:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

102:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

103:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

104:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

105:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

106:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

107:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

108:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

109:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.